

GlossBERT Classifier: Exploiting BERT capabilities in WSD Task

Antonio Gagliardi

Master Artificial Intelligence and Robotics

Sapienza, University of Rome

gagliardi.1977681@studenti.uniroma1.it

Abstract

WSD is a task aimed at assigning to one or more words their correct meaning in their surrounding context. Usually this task is treated as a Multi-Classification problem, but it is shown that it is possible to deal with the complexity of the task switching it into a Binary-Classification domain. To solve the task, we need a Language Model (LM), which contains the semantics of the desired language (in this case, English). Google’s BERT architecture (a transformer-type architecture consisting only of an Encoder) is presented and, thanks to a fine-tuning process, it is shown to be capable of solving the task even though it has not been trained to do so.

1 WSD: Gloss Knowledge Incorporation

1.1 WordNet

Thanks to WordNet, (Miller, 1994) we can access a lexical-semantic database that contains structured knowledge about the English language. WordNet organizes its knowledge into groups of synonyms called **Synsets** (from the point of view of our task, a synset may well represent a group of ambiguous words). They are refined into **Senses** (identified by a specific **Sense-Key**) that refers to a particular interpretation of an ambiguous word, followed by a **Definition**.

1.2 GlossBERT approach for WSD

As Huang et al. (2019) say, it is possible to use BERT of Devlin et al. (2019) to perform WSD tasks in a Binary Classification problem domain. We can take advantage of the properties of BERT that make it a suitable candidate for LM: Since BERT has been trained on an extended English text corpus, it has a good knowledge of the semantics of the language itself. We can exploit this and construct new sentences containing the definition for a particular word that we want to disambiguate in

context, resulting in a set of **right/wrong context-gloss pairs**. Feeding these information into the transformer, we can train a linear classifier on top of BERT that is able to predict whether a particular definition is appropriate for that particular word in context.

2 Data Processing

Huang et al., 2019 proposed two different types of sentence pre-processing to make the data more suitable for WSD using the BERT model:

- No Supervision: The original sentence (acting as **context sentence**) is concatenated with each of the possible meanings of the ambiguous word we want to disambiguate, and formatted as `context[SEP]definition`.
- Weak-Supervision: It takes the original sentence in which the ambiguous word is highlighted as "word" and then concatenates it with each possible meaning, formatted as `context[SEP]word:definition`.

Processed	Training	Validation	Test
No	12339	685	685
Yes	45513	2505	2505

Table 1: Datasets

Since every sentence $s_i \in S$ consists of words $s_i = \{w_1, w_2, \dots, w_n\}$ and some words $\{t_1, t_2, \dots, t_j\} \subset \{w_1, w_2, \dots, w_n\}$ are ambiguous, for each ambiguous word we have a set of $\{c_1, c_2, \dots, c_m\} \in C_{t_j}$ possible senses retrieved from WordNet. This means that $\forall t_j \in s_i, \forall c_m \in C_{t_j}$, s_i becomes $\{s_i c_1, s_i c_2, \dots, s_i c_m\}$ for every possible t_j in the sentence. There are ground truth senses marked as YES and possible candidate senses marked as NO based on the information contained in the `train-fine-grained.json` file.

This procedure leads also to a consistent increase in the size of our dataset, which is generally a good practice for training an effective neural network. See Table.1

3 Overall Model Architecture

The model consists of two main parts:

1. BERT: Takes the sequence of encoded context-gloss pairs tokens as input and encodes them into the latent representation of the transformer. As input to the classifier, we use the special BERT's token [CLS] from the last hidden state of the encoder, which contains the general meaning of the entire processed sentence.
2. Linear Classifier: It receives as input the features of the BERT [CLS] token and performs a Binary Classification using a Softmax.

4 Coarse-Grain WSD

We must perform our classification task using **Coarse-Grained Senses**, which are nothing more than clusters of Fine-Grained Senses with similar meanings. Since we have a coarse-fine-defs-map.json file that identifies the fine-grained senses within a coarse-grained cluster, we can use the fine-grained meanings of ambiguous words to construct our sentence-gloss pairs for neural network training. In this way, when we ask our model to predict the correct coarse-grain cluster for a particular ambiguous word, it returns the coarse-grain cluster that contains the index of a sentence-gloss pair marked as YES and identified as the most probable to disambiguate that particular word in context.

5 Training

Training was performed using Adam as Optimizer, with a learning rate of 0.00002, batch sizes of 32 with Gradient Accumulation. The latent dimension of BERT¹ is 768 and last four layers (this indication is purely empirical but usually seems to be a good option) of the transformer are defrozeed for the fine-tuning process, BERTTokenizer is used to tokenize the split context-gloss pair sets, with padding set to its maximum length of 512. The Classifier, which consists of a single output layer

¹<https://www.kaggle.com/code/rhtsingh/utilizing-transformer-representations-efficiently>

with bias, is trained with BCEWithLogitsLoss, which includes a Non-Linear Softmax Activation Function to scale the output in the range of $[0, 1]$ probability distribution. Looking at Figure.3 it's possible to see that the dataset is not so imbalanced, so we do not need to use an additional weight tensor to focus the neural network's attention on the less numerous samples.

6 Experimental Results

F1 Score Results (Training Procedure)

Pre-Processing	Training	Validation
No Supervision	0.8735	0.8636
Weak-Supervision	0.9068	0.8910

F1 Score Results (Model Testing)

	Coarse-Grain	Fine-Grain
No Supervision	0.9158	0.5231
Weak-Supervision	0.9204	0.5231

Table 2: Training procedure results and Model testing

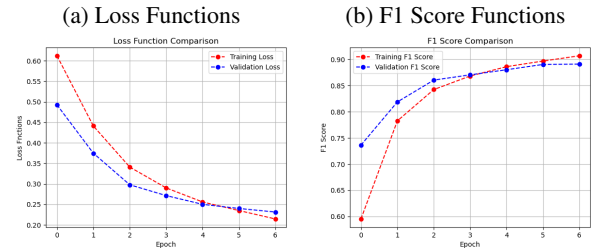


Figure 1: Loss Function and F1 Score after a few epochs of training the GlossBERT Classifier with **Weak Supervision** in data pre-processing. During the last epoch, training was stopped at epoch 6 by Early Stopping to prevent possible overfitting.

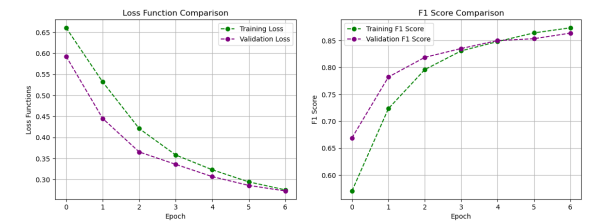


Figure 2: Loss Function and F1 Score after a few epochs of training the GlossBERT Classifier with **No Supervision** in data pre-processing. During the last epoch, training was stopped at epoch 6 by Early Stopping to prevent possible overfitting. We can also observe that the presence of no supervision leads to a lower performance in terms of accuracy of the model (as already mentioned in Table.2).

6.1 Interpretation

Results shown in Table.2 highlights better performance in the model trained with weak-supervision mechanism. One explanation for why GlossBERT performs better in the presence of weak supervision pre-processing is that we give BERT a more important semantic cue to focus on: highlighting the ambiguous target word with the symbols " " and consequently creating the gloss as [SEP]word:definition makes BERT [CLS] token more precise in terms of the overall meaning of the sentence (it seems like we incorporate the definition of the word more into the word itself).

7 Conclusions and possible steps further

GlossBERT seems to do its job quite well, but only the definition of a given ambiguous word is considered in providing further semantic information to the transformer when a **context-gloss pair** is constructed. A possible further research step could be to try a small improvement to make the gloss more precise, adding also an example of the ambiguous word given its definition. Thus, the definition of the gloss could be changed from [SEP]word:definition to [SEP]word:definition[SEP]word:example, introducing a brand new **Strong-Supervision** approach in data-processing. However, the main lexical-semantic knowledge resources such as WordNet (Miller, 1994) and BabelNet (Navigli and Ponzetto, 2010) lack examples for many words in the dataset. Due to this limitation, this approach is only theoretical and is not physically implemented in this work.

References

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- Luyao Huang, Chi Sun, Xipeng Qiu, and Xuanjing Huang. 2019. [GlossBERT: BERT for word sense disambiguation with gloss knowledge](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3509–3514, Hong Kong, China. Association for Computational Linguistics.

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A Appendix

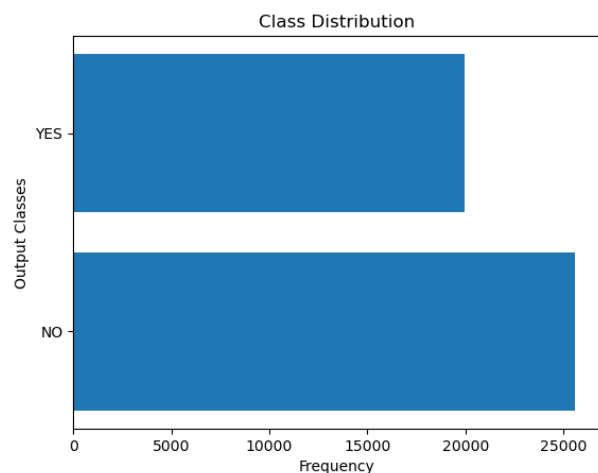


Figure 3: Classes Distribution: Note that the dataset is not so unbalanced, YES and NO constitute respectively the 0.44 and 0.56 of the data.